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- (h) Indicator used in non-aqueous acid base titration is _____.
- Starch
 - Mordant black-III
 - KMnO_4
 - Crystal violet
- (i) Formation of colored compounds at the end point in Precipitation titration is a characteristic of _____ method.
- Mohr's
 - Fajans
 - Gay-Lussac's
 - Volhard's
- (j) Purpose of masking and demasking agents is employed to _____.
- Estimate Metal
 - Heavy metals
 - Estimate Metal complexes
 - All
- (k) PPM mean _____.
- Percent per millions
 - percent parts million
 - percent purity in millions
 - parts per million
- (l) According to _____ theory acid is an electron acceptor.
- Arrhenius theory
 - Lewis theory
 - Ostwalds theory
 - lorry Bronsted theory
- (m) Color of Phenolphthalein in alkaline pH is _____.
- Red
 - Pink
 - Yellow
 - Colorless
- (n) The indicator employed in Volhards methods is _____.
- Ferric nitrate
 - Ferrioin
 - Crystal violet
 - None
- (o) Following is not reference electrode.
- Std. Hydrogen electrode
 - Calomel electrode
 - Quinhydrone electrode
 - Silver-silver chloride electrode
- (p) Ceric sulphate used as oxidizing agent is often termed as _____.
- Gravimetry
 - Permanganatometry
 - Cerimetry
 - None
- (q) _____ is popular redox self indicator employed in Redox Titrations.
- KMnO_4
 - Ferrioin
 - Starch
 - None of above
- (r) DME stands for _____ electrode.
- Diffused mercury
 - Direct measuring
 - Dropping mercury electrode
 - Diffusion measuring

(s) Diazotization titration method is suitable for assay of _____

- a) Primary aliphatic amine
- b) Primary aromatic amine
- c) Secondary aliphatic amine
- d) tertiary aromatic amine

(t) Ohm's law can be mathematically represented as _____

- a) $R \propto 1/a$
- b) $I = E/R$
- c) $C = 1/R$
- d) $R = \rho l/a$

Que. No. 2 Attempt any TWO of the following:

(10 x 2 = 20)

- a) Explain principle and steps involved in the gravimetric analysis.
- b) Explain titration curve with example of
 - i. Strong Acid Vs Strong Base
 - ii. Strong Acid Vs Weak Base
 - iii. Weak Acid Vs Weak Base
 - iv. Weak Acid Vs Strong Base
- c) What is potentiometric analysis? Enlist types of electrodes used in potentiometry. Explain construction and working of standard hydrogen electrode.

Que. No 3. Attempt any SEVEN of the following

(5 x 7 = 35)

- a) What are errors? Explain determinant and indeterminate errors.
- b) Write a note on primary and secondary standards.
- c) Explain types of solvent used in non aqueous titration with suitable example.
- d) How is Calcium Gluconate assayed? Explain preparation and standardization of Disodium EDTA.
- e) Write principle behind the diazotization titration.
- f) Classify Redox titration? Enlist the difference between Iodometric and Iodimetric titration.
- g) Give the principle involve in the permanganate titration. How KMnO_4 is standardized.
- h) Write a note on principle, construction and working of polarography.
- i) Explain the principle involve in the conductometry. Add a note on effect of dilution on conductance.

----- END OF PAPER -----

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE - RAIGAD - 402103**

Supplementary Examination, June 2018

Branch: B. Pharmacy

Semester: I

Subject with Subject Code: Pharmaceutical Analysis-I
[BP102T]

Marks: 75

Date: 06 / 06 / 2018

Time: 3 Hrs.

Instructions to Students:

- 1] All Questions are compulsory.
- 2] Figures/ structures to the right indicate full marks.

Q. 1 Choose the correct alternative.

[20]

- i. The number of gm-equivalent of the solute per liter of solution is known as
 - A. Normality
 - B. Molarity
 - C. Molality
 - D. Mole fraction
- ii. In strong acid – strong base titration, the pH of mixture at initial stage is find out by formula
 - A. $\text{pH} = -\log[\text{H}^+]$
 - B. $[\text{H}^+] = \frac{\text{NaVa} - \text{Nb Vb}}{(\text{Va} + \text{Vb})}$
 - C. $\text{POH} = -\log[\text{OH}^-]$
 - D. $[\text{OH}^-] = \frac{\text{NbVb} - \text{NaVa}}{(\text{Va} + \text{Vb})}$
- iii. Methyl orange has a pH range of
 - A. 2.2 to 4.0
 - B. 3.2 to 4.4
 - C. 4.4 to 5.4
 - D. 4.2 to 6.2
- iv. In acid base titrationelectrode is used as indicator electrode.
 - A. Platinum
 - B. Glass
 - C. Mercury
 - D. Silver
- v. One of the following statements is NOT true:
 - A. Accuracy expresses the correctness of measurement
 - B. Precision represents reproducibility of measurement
 - C. High degree of precision implies high degree of accuracy also
 - D. High degree of accuracy implies high degree of precision also
- vi. Glacial acetic acid is an example of
 - A. protogenic solvent
 - B. protophillic solvent
 - C. amphiprotic solvent
 - D. Aprotic solvent

vii. Which of the following is used as indicator in Mohr's method?

- A. Starch
- B. Ferric ammonium sulphate
- C. Phenolphthalein
- D. Potassium chromate

viii. _____ gm of AgNO_3 in 1 liter distilled water gives 1 M AgNO_3 solution.

- A. 40
- B. 169.87
- C. 76.12
- D. 83.5

ix. _____ is masking agent.

- A. Ascorbic acid
- B. Sodium cyanide
- C. Hydroxylamine
- D. All of these

x. Murexide is used for determination of _____.

- A. Ba^{++}
- B. Cd^{++}
- C. Zn^{++}
- D. Ca^{++}

xi. For gravimetry _____ filter papers are used.

- A. Ash
- B. Ash less
- C. Normal
- D. None

xii. Impurities precipitate out after precipitation of analytes called as _____.

- A. Post precipitate
- B. Co- precipitate
- C. Past precipitate
- D. None of these

xiii. If acidified Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) acts as oxidizing agent, color changes from _____.

- A. orange to red
- B. orange to green
- C. yellow to green
- D. yellow to red

xiv. In Ceriometric titration which of the following indicator is used?

- A. Ferroin
- B. Phenolphthalein
- C. Starch
- D. Potassium dichromate.

xv. In the standardization of $\text{Na}_2\text{S}_2\text{O}_3$ using $\text{K}_2\text{Cr}_2\text{O}_7$ by iodometry, the equivalent weight of is

- A. (Molecular weight)/2
- B. (Molecular weight)/6
- C. (Molecular weight)/3
- D. Same as molecular weight

xvi. Basicity of organic acids can be determined by

- A. Conductometry
- B. Refractometry
- C. Non-aqueous titrations
- D. Complexometry

xvii. The ratio of cell constant and resistance in Conductometric titration is known as

- A. EMF
- B. Specific conductance
- C. Standard potential
- D. none of these

xviii. Silver-silver chloride reference electrode is made up of

- A. copper wire coated with copper chloride
- B. Sodium wire coated with sod. chloride
- C. Mercury with calomel
- D. Silver wire coated with silver chloride

xix. In polarography any change in diffusion current is denoted by

- A. Ilkovic equation
- B. Nernst Equation
- C. Arrhenius equation
- D. Stock equation

xx. Polarography can be used for the

- A. To study fluorescent material
- B. Study of resistance of a solution
- C. Study of current potential relationship
- D. Study of optical activity of organic molecules

Q. 2 Answer any TWO of the following questions.

[20]

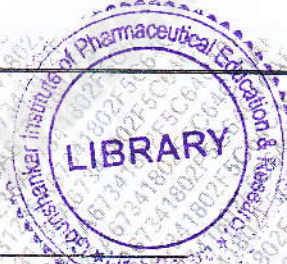
- A) Give in Detail End Point Determination in Precipitation Titrations.
- B) Explain in detail the principle involved in polarographic technique and its instrumentation.
- C) Explain in detail neutralization curves for
 - i. Strong acid vs strong base
 - ii. Strong acid vs weak base
 - iii. Weak acid vs weak base.

Q. 3 Answer any SEVEN of the following questions.

[35]

- A) Discuss in detail Iodine Titration.
- B) Write a note on Error.
- C) Discuss about different solvents used in non-aqueous acid base titration.
- D) Differentiate between Co-precipitation and post-precipitation.
- E) Discuss various types of electrode. Explain Normal Hydrogen Electrode in detail.
- F) Explain in detail limit test for Arsenic.
- G) Write a short note on various types of conductometric titration.
- H) Give the principle, method and application of Diazotization titration.
- I) Write a note on Metalochromic indicators.





Branch: B. Pharmacy

Subject / code: Pharmaceutical Analysis -I (BP 102T)

Date: 19/12/2018

Semester: I

Marks: 75

Time: 3 Hours

Instructions: i) All questions are compulsory

ii) Figures to the right indicate full marks

iii) Draw the diagrams or flow charts wherever necessary.

Q. No. 1 Attempt the following multiple choice questions. (1 X 20=20)

- Which one the following compound is assayed by gravimetric analysis
 - ceric sulphate
 - barium sulphate
 - calcium gluconate
 - Benzoic acid
- Reduction is ----- of electrons.
 - loss
 - gain
 - removal
 - none of these
- Gram----- of solute dissolved in 1 liter of solvent gives 1 Normal solution
 - equivalent Weight
 - molecular Weight
 - Mole
 - none of this
- From the following ----- is not a self indicator?
 - ferroin
 - iodine
 - potassium permanganate
 - ceric sulphate
- Colour of phenolphthalein in alkaline pH is -----
 - yellow
 - brown
 - pink
 - red
- PH is given as -----
 - $\text{pH} = \text{Log} [\text{H}^+]$
 - $\text{pH} = -\text{Log}_{10} [\text{H}^+]$
 - $\text{pH} = 1/[\text{H}^+]$
 - $\text{pH} = -\text{Log}_{10} [\text{OH}^-]$
- Drop of mercury in DME acts as ----- electrode.
 - reference
 - indicator
 - hydrogen
 - calomel

8. -----ml of hydrochloric acid is diluted with 1 liter of water to make 1M solution.
- A) 36.5 B) 85
C) 8.5 D) 3.65
9. ----- of the following is hexadentate ligand.
- A) glycine B) ethylene diamine
C) ammonia D) EDTA
10. Solubility of precipitate is-----on addition of common ions.
- A) increased B) decreased
C) remains constant D) none of these
11. In Volhard's method -----is used as titrant.
- A) potassium chromate B) ferric ammonium sulphate
C) ammonium thiocyanate D) silver nitrate
12. According to Lewis acid is a substance which-----
- A) accept proton B) accept electron pair
C) donate proton D) donate electron pair
13. Benzene is -----
- A) protophilic solvent B) protogenic solvent
C) aprotic solvent D) amphiprotic solvent
14. Potential of standard hydrogen electrode is-----Volt.
- A) 0.245 B) 0.385
C) 0.199 D) 0.0
15. ----- is the requirement of primary standard substances.
- A) purity B) stability
C) no water of hydration D) all of these
16. The -----states that "The rate of chemical reaction is proportional to the active masses of reacting substances".
- A) law of mass action B) first law of thermodynamics
C) second law of thermodynamics D) universal law
17. Iodometry deals with titration of -----in chemical reaction.
- A) iodine liberated B) standard solution of iodine
C) both A and B D) none of these
18. In Potentiometry, electrode whose potential remains constant is called as -----.
- A) anode B) indicator electrode
C) reference electrode D) inert electrode

19. Current carrying capacity of ions is nothing but-----.

- A) voltage B) conductance
C) potential difference D) ampere

20. Ilkovic equation is given by the formula -----

- A) $607 \text{ n CD}^{1/2} \cdot \text{m}^{2/3} \cdot \text{t}^{1/6}$ B) $307 \text{ n CD}^{1/2} \cdot \text{m}^{2/3} \cdot \text{t}^{1/6}$
C) $700 \text{ n CD}^{1/2} \cdot \text{m}^{2/3} \cdot \text{t}^{1/6}$ D) $706 \text{ n CD}^{1/2} \cdot \text{m}^{2/3} \cdot \text{t}^{1/6}$

Q. No. 2 Attempt any two of the following

(10 X 2=20)

1. Explain principle of Conductometric titration and explain Conductometric titration curve with example for
 - i) Strong acid Vs strong base
 - ii) weak acid Vs strong base
 - iii) Strong acid Vs weak base
 - iv) weak acid Vs weak base
2. Describe in detail 'Theories of acid base Indicators'
3. Explain in detail various steps involved in gravimetric analysis.

Q. No. 3 Attempt any seven of the following

(5X 7=35)

1. Define Errors; explain its types and method of minimizing determinant errors.
2. Explain types of solvents used in non-aqueous titration with suitable examples.
3. How will you prepare i) 0.1N sodium hydroxide ii) 0.5 N sodium thiosulphate
4. Explain in detail principle and mechanism of diazotization titrations.
5. Differentiate between Iodimetry & Iodometry.
6. Enlist different methods of precipitation titrations. Explain mohl's method.
7. Explain concept of oxidation & reduction with example.
8. Explain principal & mechanism of metal ion indicators.
9. What is reference electrode? Write construction and working of standard hydrogen electrode

----- END OF PAPER -----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Semester Examination – May - 2019

Branch: B. Pharm

Sem.:- I

Subject with Subject Code:- Pharmaceutical Analysis – I
(BP102T)

Marks: 75

Date:- 15 -05-2019

Time:- 3 Hr.

Instructions to the Students

1. All questions are compulsory.
2. Figures to the **right** indicate full marks.
3. Attempt all Multiple Choice Questions in the main answer book.

(Marks)

Q.1. Attempt the following multiple choice question

(20 x 1 = 20)

1) The difference between the residual current and limiting current is called ascurrent.

- | | |
|--------------|------------------|
| a) Migration | b) Diffusion |
| c) Kinetic | d) None of these |

2) is Self indicator

- | | |
|--------------------|--------------------|
| a) HClO_4 | b) KMnO_4 |
| c) AgNO_3 | d) All |

3) Metallochromic indicators used in

- | | |
|----------------------------------|-----------------------------|
| a) Argentometric titration | b) Complexometric titration |
| c) Oxidation Reduction titration | d) Gravimetric titration |

4) EDTA is

- | | |
|----------------|------------------|
| a) Monodentate | b) Bidentate |
| c) Hexadentate | d) None of these |

5) Which of the following is aprotic solvent

- | | |
|--------------------|----------------|
| a) Ethanol | b) Benzene |
| c) Perchloric acid | d) Acetic acid |

6) According totheory the undissociated indicators has colour different than its ion

- a) Resonance
- b) Chromophore
- c) Ostwald
- d) Univarsal

7) Buffer solution is that solution that

- a) Resistance to change in colour
- b) Resistance to change in concentration
- c) Resistance to change in pH
- d) Resistance to change in potential

8) In gravimetry, digestion gives

- a) Small crystals
- b) Pure precipitate
- c) Impure precipitate
- d) None of these

9) Which of the following electrode is in the depolarised from?

- a) Indicator
- b) Reference
- c) Micro
- d) All of them

10) The conductance is reciprocal of.....

- a) Resistance
- b) Specific Resistance
- c) Specific conductance
- d) Equivalent conductance

11) Precipitation of colloidal particles is.....

- a) Peptisation
- b) Coagulation
- c) Digestion
- d) Ignition

12) What is the reagent used for diazotization?

- a) NaNO_2 + dilute HCl
- b) KNO_3 + dilute H_2SO_4
- c) Zn + dilute H_2SO_4
- d) Tin + dilute H_2SO_4

13) Which of the following is primary standard?

- a) As_2O_3
- b) $\text{K}_2\text{Cr}_2\text{O}_7$
- c) KIO_3
- d) All of them

14) Example of strong acid & weak base salt is.....

- a) NaCl
- b) KCl
- c) KCN
- d) NH_4Cl

15) The unit of current is.....

- a) Amperes
- b) Volts
- c) Ohm
- d) Mhos

16) Which of the following indicator is used for assay of sodium benzoate?

- a) Crystal Violet
- b) Phenol Red
- c) Phenolphthalein
- d) None of these

17) Assay of magnesium sulphate is type of titration.

- a) Redox
- b) Acid base indicator
- c) Complexometric
- d) Non-aqueous

18) Pool of mercury is called as.....

- a) Indicator electrode
- b) Reference electrode
- c) Micro electrode
- d) None of them

19) Which of the following is not a Complexometric indicator?

- a) Catechol violet
- b) Crystal Violet
- c) Mordant black 17
- d) Mordent Red 7

20) Which of the following is not a step of gravimetry analysis?

- a) Filtration of precipitate
- b) Washing of precipitate
- c) Ignition of precipitate
- d) End point detection

Q.2. Attempt any TWO of the following:

(2 x 10 = 20)

- 1) What is gravimetric analysis? Explain in detail the step involved in gravimetric analysis.
- 2) What is acid base titration? Explain neutralization curves with suitable example.
- 3) Explain the principle behind potentiometer. Write a construction and working of reference electrode and indicator electrode

Q.3. Attempt any SEVEN of the following:**(7 x 5 = 35)**

- 1) What are errors? Enlist sources of errors and explain types of errors with suitable example.
- 2) Define the terms
 - a) Qualitative analysis
 - b) Normality
 - c) Titrant
 - d) Primary standards
 - e) Indicator
- 3) Explain Ostwald theory and Resonance theory of acid base indicators.
- 4) What is non-aqueous titration and explain different types of non-aqueous solvents with examples.
- 5) Explain oxidation and reduction with suitable example? Write a short note on Iodimetry with suitable examples.
- 6) What is complexometric titration? Write in detail complex formation & its stability
- 7) What is precipitation titration? Write in detail about Mohr's method.
- 8) What is standardization? How to standardize potassium permanganate and ceric ammonium sulphate
- 9) Explain principle behind polarography. Write a construction and working of polarography.

END



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination - December - 2019

Course: B.Pharm.

Subject: Pharmaceutical Analysis-I (BP102T)

Date: 27/12/2019

Semester: I

Marks: 75

Time: 3hrs

- Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary.

Q.No.1 Attempt the following questions (All Questions Compulsory)

(20 Marks)

- Concentration is not expressed is -----
 - w/v
 - v/w
 - v/v
 - w/w
- Ionic product of water is -----
 - 1×10^{-13}
 - 1×10^{-14}
 - 1×10^{-15}
 - 1×10^{-16}
- Which of the following is primary standard substance?
 - Oxalic acid
 - Sodium Hydroxide
 - Potassium Permanganate
 - All of Above
- Indeterminate error is also called as -----
 - Operational error
 - Instrumental error
 - Error of method
 - Random error
- How many grams of Sodium hydroxide will be required to prepare 250 ml of 0.1N solution?
 - 10
 - 1
 - 0.1
 - 0.01
- What is color change interval of methyl Red (pH Change)?
 - 0.8 - 1.2
 - 1.2 - 2.8
 - 2.8 - 3.6
 - 4.2 - 6.3
- Number of moles of solute per 1000 gm of solvent gives ----- solution.
 - 1 Molar
 - 1 Molal
 - 1 Normal
 - 1 Percent
- In titration of weak base with weak acid, the solution at equivalence point will be -----
 - Neutral
 - Acidic

- C. Basic
D. No change in pH
9. Methanol is ----- solvent used in Non-aqueous titration.
A. Aprotic
B. Protogenic
C. Protophilic
D. Amphiprotic
10. The indicator employed in Volhards method is -----.
A. Ferric ammonium sulphate
B. Ferroin
C. Crystal violet
D. None of these
11. Dead stop end point method is also known as -----.
A. Potentiometry
B. Conductometry
C. Titration with 2 indicator electrodes
D. Titration with indicator and reference electrode
12. The conductance that changes on dilution is ----- conductance.
A. Molar
B. Equivalent
C. Specific
D. All of these
13. Drop of mercury in DME acts as ----- electrode.
A. Reference
B. Indicator
C. Hydrogen
D. Calomel
14. Titrations with standard solution of Iodine are termed as -----.
A. Iodometry
B. Iodimetry
C. Redox
D. All of these
15. EDTA is popularly employed as a -----.
A. Multidentate ligand
B. Complexing agent
C. Chelating agent
D. All of these
16. Ferrous sulphate is assayed by ----- titration.
A. Ceriometry
B. Iodimetry
C. Mercurimetry
D. Bromatometry
17. Which of the following Titrant is act as self-indicator?
A. Potassium Permanganate
B. Silver Nitrate
C. Sodium Hydroxide
D. Hydrochloric Acid
18. Diazotization method is carried out for following compound -----.
A. Aromatic Amine
B. Phenol
C. Aromatic Aldehyde
D. Aromatic Ketone

19. Glass electrode is example of ----- electrode.

- A. Reference
- B. Ion selective
- C. Indicator
- D. Both B & C

20. What is emf -----.

- A. Electro motive force
- B. Electron motion force
- C. Electro motion force
- D. All of these

Q.No.2 Attempt any TWO questions of the following: (20 Marks)

- A] Explain different types of Acid-Base titration; add a note on neutralization curve.
- B] Explain various steps involved in gravimetric analysis.
- C] Give the theory behind conductometry. Explain different shapes of graphs obtained during acid-base titrations. Define different conductances and add note on effect of dilution on conductances.

Q.No.3 Attempt any SEVEN questions of the following: (35 Marks)

- A] Write a note on theories of indicator.
- B] Explain types of errors and methods to minimize the errors.
- C] Write a note on Diazotization method.
- D] Differentiate between Mohr's and Volhard's method.
- E] Write a note on different types of Complexometric titration.
- F] Write in details about solvents used in non-aqueous titrations with appropriate examples.
- G] Explain principle involved in conductometric titrations and give its applications.
- H] Write short note on reference electrodes.
- I] Give basic principle of Polarography. Explain construction and working of DME with its advantages and disadvantages.

----- END OF PAPER -----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 10/05/2023

Course	: B. Pharmacy	Sem	: I
Subject Name	: Pharmaceutical Analysis-I	Subject Code	: BP102T
Max Marks	: 75	Duration	: 3 Hr

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) What is oxidizing agent?
- ii) What base according to Lowery-Bronsted theory?
- iii) Why potentiometer is also called as pH meter?
- iv) Enlist names of limit tests official in Indian Pharmacopeia.
- v) What is mean by primary standard?
- vi) Give the example of self-indicator and external indicator.
- vii) What is mean by equivalent conductance?
- viii) What is masking agent?
- ix) Define Molarity.
- x) What is diazotization titration?

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) What is Gravimetry? Explain the steps involved in Gravimetric Analysis.
- ii) Describe the concept conductometric titrations. Explain in detail different types of conductometric titrations along with one example.
- iii) What is mean by redox titration? Enlist its types and types of indicators used in redox titration. Write in detail about Iodine titrations.

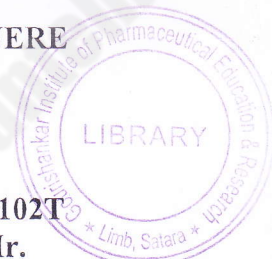
Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) What is impurity? Explain sources of impurities in medicinal agents.
- ii) Differentiate between Mohr's and Volhard's method.
- iii) What is error? Explain in brief types of errors.
- iv) What is potentiometry? Explain types of electrodes used in it along with one example.
- v) What are titration curves? Write in brief various acid-base titration curves.
- vi) Write the principle involved in polarography. Add a note on DME.
- vii) Define Indicator. Add a note on theories of indicators.
- viii) What is non-aqueous titration? Write a note on properties and classes of solvents used in it with example.
- ix) What is complexometric titration? Explain different classes of complexometric titration along with one example.

-----END OF THE PAPER-----

Course : B. Pharmacy
Subject Name : Pharmaceutical Analysis-I
Max Marks : 75

Sem: I
Subject Code : BP102T
Duration : 3 Hr.

**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define primary & secondary standard substances with an example of each.
- ii) What do you mean by oxidizing and reducing agent?
- iii) Give the principle of limit test of chloride.
- iv) Write down the definition of acid and base as per Lowry-Bronsted theory?
- v) What is levelling and differentiating effect?
- vi) What is masking and demasking?
- vii) How will you standardize 0.1N KMnO₄ Solution?
- viii) Write down the principle of Volhard's method.
- ix) Define equivalent conductance.
- x) Give the principle of diazotization titration?

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Describe the principle of Potentiometric titration. Write down the construction and working of indicator electrode and reference electrode.
- ii) Explain different types of acid base titrations. Add a note on neutralization curve.
- iii) Define gravimetric analysis. Describe the various steps involved in it.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) What is Pharmaceutical Analysis? Give the classification of different techniques of analysis.
- ii) Define error. Enlist the sources and types of errors and explain methods of minimizing errors.
- iii) What do you mean by impurity? Describe the sources of impurities in pharmaceuticals.
- iv) Define indicator and explain Ostwald theory and resonance theory of acid base indicators.
- v) What is non aqueous titration? Explain different types of non-aqueous solvents with examples.
- vi) What is precipitation titration? Write in detail about Mohr's method.
- vii) Explain the concept of redox titration with suitable example and differentiate between Iodimetry and Iodometry.
- viii) Write a short note on Conductometric titrations.
- ix) Explain the principle of Polarography. Focus on construction and working of dropping of mercury electrode and rotating platinum electrode.

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2023

Date: 11/12/2023

Course	:	B. Pharmacy	Sem	:	I
Subject Name	:	Pharmaceutical Analysis I	Subject Code:		BP102T
Max Marks	:	75	Duration	:	3 Hr.

Instructions:

- 1. All questions are compulsory**
- 2. Draw diagrams / figures wherever necessary**
- 3. Figures to right indicate full marks**

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- Explain principle of Mohr's method.
- Explain principle, reaction involved in standardization of sodium hydroxide.
- Differentiate between iodometry and iodimetry.
- Define co-precipitation and post-precipitation.
- Explain back titration with an example
- Define reference electrode and indicator electrode with example.
- Describe principle of polarography.
- Define oxidizing and reducing agent with example.
- Differentiate between qualitative and quantitative analysis.
- Differentiate between accuracy and precision.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- Describe in detail conductometric titrations.
- Explain principle and different steps involved in Gravimetric analysis?
- Write the principle of limit test for arsenic? Explain construction and working of Gutzeit test apparatus.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- Give the principle, reaction for limit test for chloride and sulphate.
- Explain in short masking and demasking agents.
- Explain the principle, reaction involved in the estimation of Calcium Gluconate.
- Describe leveling and differentiating effects in the context of non-aqueous titrations.
- Discuss construction and working of calomel electrode.
- Define and classify errors.
- Discuss neutralization curve of strong acid against strong base.
- Differentiate between Volhard's and modified Volhard's method.
- Explain pM indicators.

-----END OF THE PAPER-----

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Supplementary Summer Examination – 2024**

Course: B. Pharmacy

Semester: II

Subject Name: Pharmaceutical Analysis I

Subject Code: BP102T

Max Marks: 75

Date: 24-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) What are the effects of impurities on pharmaceutical products?
- ii) Define Lewis Acid and Base with a suitable example.
- iii) Why buffers are added in complexometric titrations.
- iv) What are the different analytical techniques used in pharmaceutical analysis?
- v) Name the indicators used in K. Fajan's method of precipitation titration.
- vi) Write a short note on demasking agents used in complexometric titrations.
- vii) Give construction of Guitzet apparatus.
- viii) Define the terms Molarity and Normality.
- ix) Explain different sources of impurities.
- x) Give the difference between Cerimetry and Iodimetry.

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Give a brief account of the limit test of Arsenic with principle, apparatus and procedure.
- ii) Give a detail account of various end point detection techniques in precipitation titration.
- iii) Explain different types of complexometric titrations.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Discuss in detail limit test for iron.
- ii) Write note on the co-precipitation process.
- iii) Give in detail the different techniques of filtration used in gravimetric analysis.
- iv) Give a brief account of leveling effect in non-aqueous titrations.
- v) Write a note on construction and working of Standard hydrogen electrode.
- vi) Explain Mohr's and Volhard's end point detection methods.
- vii) Give the construction and working of dropping mercury electrode.
- viii) Explain the construction of different types of conductometric cells.
- ix) Discuss in brief the factors affecting surface adsorption.

***** END OF THE PAPER *****

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